

Arunesh Vijayanand

+91 91766 17573 | aruneshvijay2703@gmail.com | linkedin.com/in/arunesh-v | github.com/ARUN3SH-27

SUMMARY

A results-driven engineering professional with hands-on experience across backend development, ML pipelines, and data analytics. Proven ability to take projects end-to-end — from raw data ingestion and model building to API deployment and stakeholder reporting. Delivered a fake-account detection API achieving 99.65% accuracy and a city-level market segmentation pipeline with F1: 0.938 across 252 cities. Scopus-indexed published researcher on hybrid deep learning for AQI forecasting ($R^2 = 0.948$). Proficient in Python, SQL, Flask, scikit-learn, MongoDB and more. Strong analytical thinking with the ability to translate complex data into clear, actionable insights.

EXPERIENCE

Technology Associate Intern — Quantis Sphere LLP

Feb 2026 – Present

- Multi-source data ingestion pipelines in Python consuming U.S. Census, ATTOM, Zillow, Redfin, and Realtor.com APIs across 50+ markets; handled pagination, rate limits, schema normalisation, and deduplication.
- GMM clustering + Logistic Regression pipeline (scikit-learn) classifying 252 Ohio cities into 4 investment tiers (F1: 0.938); modular preprocessing, feature engineering, and reproducible output artefacts.
- OLS regression models (statsmodels/pandas) estimating CPM and CPL ranges across 4 audience clusters for a Meta Ads real estate campaign; outputs fed directly into client targeting strategy.
- Structured reporting packages (scoring model docs, data-source logs, targeting decks) translating complex datasets into actionable matrices for non-technical stakeholders.

PROJECTS

Fauxect — Fake Account Detection API | Capstone · IEEE Access submission in progress · github.com/ARUN3SH-27/Major-project

- Flask REST API serving a soft-voting ensemble (Random Forest + Gradient Boosting + Logistic Regression) with 99.65% accuracy, AUC = 0.9966 on 2,818 labelled Twitter accounts (Cresci 2015 benchmark).
- Dual-mode architecture: static endpoint for reproducible batch scoring + dynamic endpoint with live Twitter API integration, bearer token rotation, and MongoDB-based response caching with TTL expiry.
- Integrated MongoDB for account data storage and API response caching; schema supports flexible attribute ingestion across static and real-time modes.

LRX — AQI Forecasting Pipeline | Published · Emerging Science Journal, Scopus · [DOI: 10.28991/ESJ-2025-09-05-010](https://doi.org/10.28991/ESJ-2025-09-05-010)

- Hybrid LSTM + Random Forest + XGBoost ensemble for multi-city AQI forecasting; achieved $R^2 = 0.948$, RMSE = 0.014, outperforming all standalone baselines.
- End-to-end pipeline: raw sensor ingestion → preprocessing → ensemble inference → interactive Streamlit dashboard with city selection and live visualisation.

Multi-Modal Insurance Claim Verification Pipeline | HackerRank Hackathon Submission

- NVIDIA NIM vision-language model APIs orchestrated for multi-modal document + image analysis; automated claim triage, structured JSON output, and fraud flag generation.
- Implemented async requests, response parsing, on-disk caching, and ThreadPoolExecutor concurrency for parallel claim processing.

Attendees — Automated Attendance System | InisioHacks 2023 — 1st Place

- GUI-based Python application automating check-in and report generation; reduced manual processing time by 40%. Awarded 1st place at InisioHacks 2023.

TECHNICAL SKILLS

Languages: Python (primary) | SQL | JavaScript (basic) | Bash | R (basic)

Backend & APIs: Flask, REST API design, JSON serialisation, request validation, token-based auth, rate limiting, caching, API integration (Twitter, Zillow, ATTOM, Realtor.com, Meta Ads)

Data & ML: scikit-learn, XGBoost, Random Forest, Gradient Boosting, LSTM (Keras/TensorFlow), GMM Clustering, OLS Regression, Feature Engineering, Model Evaluation (AUC, F1, R^2 , RMSE)

Databases: MongoDB, SQLite, SQL (joins, window functions, CTEs, aggregations), pandas

Visualisation & BI: Streamlit, Matplotlib, Seaborn, Plotly, Power BI (basic), Google Sheets / Excel

Tools & Platforms: Git / GitHub, Jupyter, VS Code, NVIDIA NIM, Meta Ads Manager, Prompt Engineering

EDUCATION

B.Tech, Computer Science and Engineering — SRM University, Chennai

2021–2025

CGPA: 8.51 / 10.0

CERTIFICATIONS & ACHIEVEMENTS

- Machine Learning Specialization — [Coursera](#) (Andrew Ng) | [AWS Academy ML Foundations](#) | [McKinsey Forward Program 2025](#)
- InisioHacks 2023 — 1st Place | Boost Hacks II — Best Gaming Hack